

Limits of sequences — 5.1

Definition: We say a sequence (a_n) converges to a real number L if the sequence becomes arbitrarily close to L as n increases.

In this case, we write $\lim_{n \rightarrow \infty} a_n = L$.

Usual limit rules apply: If (a_n) and (b_n) converge,

then $\lim_{n \rightarrow \infty} a_n \pm b_n = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$

$$\lim_{n \rightarrow \infty} a_n b_n = \left(\lim_{n \rightarrow \infty} a_n \right) \left(\lim_{n \rightarrow \infty} b_n \right)$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n} \quad \text{as long as } \lim_{n \rightarrow \infty} b_n \neq 0$$

$$\lim_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} b_n \quad \text{whenever } a_n \leq b_n \text{ for all } n$$

$$(a) \quad (1, 1, 1, 1, 1, \dots)$$

$$a_n = 1 \rightarrow 1$$

$$(b) \quad (1, 2, 3, 4, 5, \dots)$$

$$a_n = n \rightarrow \infty$$

$$(c) \quad (92, \sin(1), 1, e^{2001}, \dots)$$

$$a_n = ? \text{ diverges (?)}$$

$$(d) \quad (1, 1, 2, 3, 5, 8, 13, \dots)$$

$$a_n = a_{n-1} + a_{n-2} \rightarrow \infty$$

$$(e) \quad (1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots)$$

$$a_n = \frac{1}{n} \rightarrow 0$$

$$(f) \quad (\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots)$$

$$a_n = \frac{n}{n+1} \rightarrow 1$$

$$(g) \quad (1, -1, 1, -1, 1, \dots)$$

$$a_n = (-1)^{n-1} \text{ diverges}$$

$$(h) \quad (3, 3.1, 3.14, 3.141, \dots)$$

$$a_n = \frac{\lfloor 10^{n-1} \pi \rfloor}{10^{n-1}} \rightarrow \pi$$

Example: $\lim_{n \rightarrow \infty} \sqrt{n+1} - \sqrt{n} = \lim_{n \rightarrow \infty} \sqrt{n+1} - \sqrt{n} \cdot \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}}$

$$= \lim_{n \rightarrow \infty} \frac{n+1 - n}{\sqrt{n+1} + \sqrt{n}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1} + \sqrt{n}} = 0$$

Squeeze Theorem: If $a_n \leq b_n \leq c_n$ for all n
and $\lim_{n \rightarrow \infty} a_n = L = \lim_{n \rightarrow \infty} c_n$, then b_n converges
to L also.

Example: $\lim_{n \rightarrow \infty} \frac{\sin(n)}{n}$, $|\sin(n)| \leq 1$
 $-1 \leq \sin(n) \leq 1$

$-\frac{1}{n} \leq \frac{\sin(n)}{n} \leq \frac{1}{n}$, and we know $-\frac{1}{n}$ and $\frac{1}{n}$
converge to 0.

So by the Squeeze Theorem,

$$\lim_{n \rightarrow \infty} \frac{\sin(n)}{n} = 0.$$

Theorem: If $a_n = f(n)$ for a function f of a real variable with $\lim_{x \rightarrow \infty} f(x) = L$,
then $\lim_{n \rightarrow \infty} a_n = L$.

This allows us to use L'Hôpital's Rule even though it doesn't make sense to take a derivative of a sequence.

Example: $\lim_{n \rightarrow \infty} \frac{\ln(n)}{n}$, $f(x) = \frac{\ln(x)}{x}$

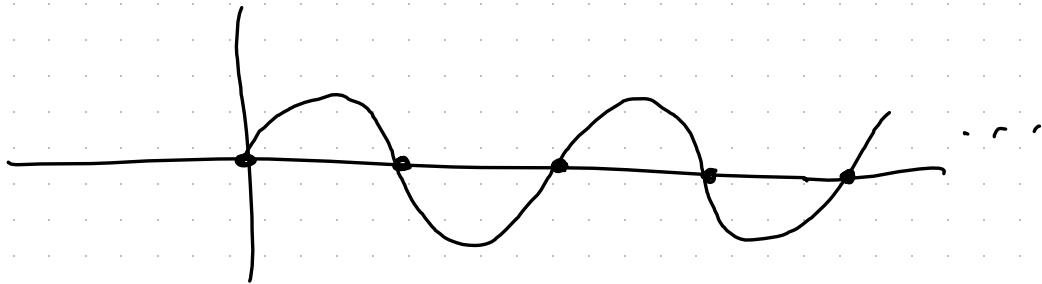
$$\text{and } \lim_{x \rightarrow \infty} \frac{\ln(x)}{x} \stackrel{2/4}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\ln(n)}{n} = 0.$$

Converge is false — just because $\lim_{n \rightarrow \infty} f(n)$ exists doesn't mean $\lim_{x \rightarrow \infty} f(x)$ does.

For instance, $\lim_{n \rightarrow \infty} \sin(\pi n) = \lim_{n \rightarrow \infty} 0 = 0$

but $\lim_{x \rightarrow \infty} \sin(\pi x)$ DNE.



Example: $\lim_{n \rightarrow \infty} \frac{e^n}{n!}$

$$\frac{e^n}{n!} = \frac{e \cdot e \cdot e \cdots e \cdot e \cdot e}{n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1} = \underbrace{\frac{e}{n} \cdot \frac{e}{n-1} \cdot \frac{e}{n-2} \cdots \frac{e}{3} \cdot \frac{e}{2} \cdot \frac{e}{1}}_{< 1}$$

So $0 \leq \frac{e^n}{n!} \leq \frac{e}{n} \cdot \frac{e}{2} \cdot \frac{e}{1} = \frac{e^3}{2n}$

$\lim_{n \rightarrow \infty} 0 = 0$, $\lim_{n \rightarrow \infty} \frac{e^3}{2n} = 0$, so by the

Squeeze theorem, $\lim_{n \rightarrow \infty} \frac{e^n}{n!} = 0$.

$n!$ grows faster than e^n

Example: $\lim_{n \rightarrow \infty} \frac{n^n}{n!}$

$$\frac{n^n}{n!} = \frac{\overbrace{n \cdot n \cdot n \cdots n}^n}{n \cdot n \cdot 1 \cdot n \cdot 2 \cdots 1} = \frac{n}{n} \cdot \frac{n}{n-1} \cdot \frac{n}{n-2} \cdots \frac{n}{2} \cdot \frac{n}{1} \gg n$$

and $\lim_{n \rightarrow \infty} n = \infty$, so by the Squeeze

theorem (or a version of it) $\lim_{n \rightarrow \infty} \frac{n^n}{n!} = \infty$

n^n grows faster than $n!$